Module II — Theory of solids

Derivations, step by step

Drude conductivity $\sigma = ne^2\tau/m$

4–5 marks

n: free electrons per unit volume. τ: mean free time between collisions (relaxation time). v_d: drift velocity. $\mathcal{E}$: applied field. J: current density.

BEFORE YOU START: WHAT THIS USES

n	free electrons per unit volume: the Drude–Lorentz picture
τ	relaxation time, the mean time between collisions: defined and averaged in the steps below
$m\,dv/dt = -e\mathcal{E}$	Newton's second law with the force on a charge $-e$ in a field $\mathcal{E}$
$J = \sigma\mathcal{E}$	Ohm's law in local form: drift velocity and mobility

START

$$m\frac{dv}{dt} = -e\mathcal{E}$$

RESULT

$$\sigma = \frac{ne^2\tau}{m}$$

FORCE AND ACCELERATION

$$m\frac{dv}{dt} = -e\mathcal{E}$$

Between collisions a free electron is pushed only by the field. Its random thermal motion averages to zero, so only this push gives a net drift.

$$v(t) = -\frac{e\mathcal{E}}{m}\,t$$

Speed gained a time t after the last collision

After a collision the electron restarts in a random direction, so its velocity averages to zero at that moment, and then it accelerates uniformly for a time t.

$$\langle t\rangle = \int_0^\infty t\,\frac{e^{-t/\tau}}{\tau}\,dt = \tau$$

Average the time since the last collision

Collisions are random, so the chance of going a time t without one is $e^{-t/\tau}$; this is what "mean time between collisions τ" means. By parts, $\int_0^\infty te^{-t/\tau}dt = \tau^2$.

$$v_d = -\frac{e\mathcal{E}}{m}\tau$$

substitute: $t \to \langle t\rangle = \tau$

The average extra velocity is $-e\mathcal{E}\tau/m$.

CURRENT DENSITY

$$J = -nev_d$$

Charge $-e$, n electrons per unit volume, moving at v_d

Check. n electrons per unit volume, charge $-e$, drift velocity v_d. What is the current density?

- ✓ $J = -nev_d$
- $J = nev_d^2$
- $J = -ev_d/n$
- $J = nmv_d$

In a time dt every electron within a distance $v_d\,dt$ of a surface of area A crosses it. They fill a box of volume $Av_d\,dt$ holding $nAv_d\,dt$ electrons of charge $-e$, so the current per area is $-nev_d$.

$$J = \frac{ne^2\tau}{m}\,\mathcal{E}$$

substitute: $v_d = -e\mathcal{E}\tau/m$

The two minus signs cancel: electrons drift against the field, but the current flows along it.

OHM'S LAW

$$\sigma = \frac{ne^2\tau}{m}$$

Compare with Ohm's law $J = \sigma\mathcal{E}$

$J \propto \mathcal{E}$ is Ohm's law, a success of the model. The mobility is $\mu = v_d/\mathcal{E} = e\tau/m$, so $\sigma = ne\mu$. Failures (three successes, four failures): the wrong temperature dependence of resistivity, and an electronic specific heat of $\frac{3}{2}R$ per mole that is not seen (why it is so small).

Free electrons in a 1-D box: E_n and E_F

5 marks

A line of length L with infinite walls at $x = 0$ and $x = L$; $V = 0$ inside. E_n: energy of level n. N: number of electrons. n_F: the highest filled level.

BEFORE YOU START: WHAT THIS USES

ψ	wavefunction of the electron, $\lvert\psi\rvert^2$ is the chance of finding it: the 1-D free-electron model
$\hbar = h/2\pi$	reduced Planck constant
$E = \hbar^2k^2/2m$	kinetic energy with de Broglie's $p = \hbar k$: shown in the first steps
Pauli	at most two electrons (opposite spins) in each level: stated at the step where it is used

START

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

RESULT

$$E_F = \frac{h^2 N^2}{32 m L^2}$$

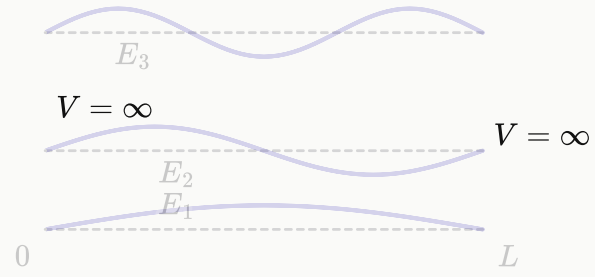
WHERE THE EQUATION COMES FROM

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

substitute: $p = \hbar k$

Inside the box the electron has only kinetic energy. A matter wave of wave number k carries momentum $p = \hbar k$ (de Broglie: $p = h/\lambda$ and $k = 2\pi/\lambda$).

$V = 0$ inside, $V = \infty$ at the walls



$$\frac{d^2\psi}{dx^2} = -k^2\psi$$

A wave of wave number k obeys this

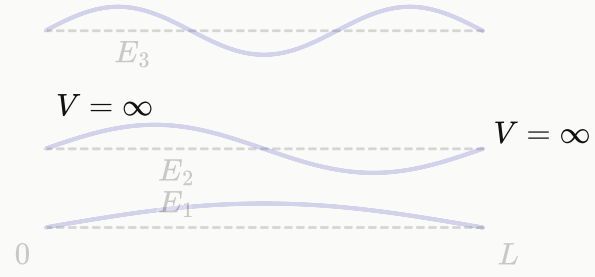
Differentiating $\sin kx$ or $\cos kx$ twice gives back $-k^2$ times the same function.

SCHRÖDINGER EQUATION INSIDE

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0$$

Put $k^2 = 2mE/\hbar^2$ from the first line into $\psi'' = -k^2\psi$. This is the time-independent Schrödinger equation with $V = 0$: only the kinetic term is left inside the box.

$V = 0$ inside, $V = \infty$ at the walls



$$\psi = A\sin kx + B\cos kx$$

Solution of $\psi'' + k^2\psi = 0$

with $k^2 = 2mE/\hbar^2$.

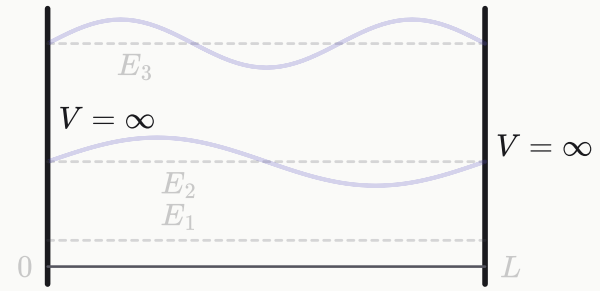
BOUNDARY CONDITIONS

$$\psi(0) = 0 \;\Rightarrow\; B = 0$$

evaluate: $x = 0$

The walls are infinitely high, so ψ must vanish at $x = 0$ and $x = L$. At $x = 0$, $\sin 0 = 0$ and $\cos 0 = 1$.

ψ must vanish at both walls



$$\sin kL = 0$$

evaluate: $x = L$

Check. $\psi(L) = A\sin kL = 0$ with $A \neq 0$. What does this force?

- ✓ $kL = n\pi, n = 1, 2, 3, \ldots$
- $kL = (n + \frac{1}{2})\pi$
- $k = 0$
- $kL = 2n\pi$ only

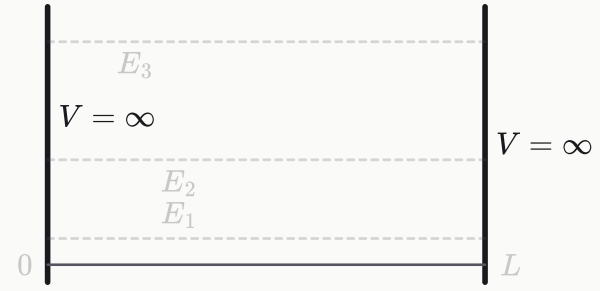
The sine is zero at whole multiples of π. $n = 0$ would give $\psi = 0$ everywhere, so n starts at 1.

$$k = \frac{n\pi}{L}$$

rearrange

Only standing waves with a whole number of half-wavelengths fit in the box.

Half-wavelengths: $n = 1, 2, 3$



NORMALISE

$$\int_0^L A^2 \sin^2 \frac{n\pi x}{L}\, dx = \frac{A^2 L}{2} = 1$$

The electron is somewhere in the box: total probability 1

$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$. The cosine part integrates to zero over whole half-wavelengths, leaving $\frac{1}{2}L$.

$$\psi_n = \sqrt{\frac{2}{L}}\,\sin\frac{n\pi x}{L}$$

Solve for A: $A = \sqrt{2/L}$

ALLOWED ENERGIES

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$

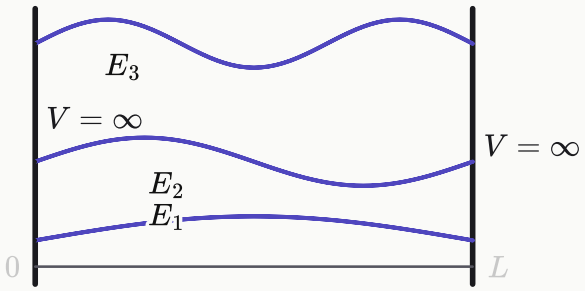
substitute: $k = n\pi/L$

$$E_n = \frac{n^2 h^2}{8mL^2}$$

substitute: $\hbar = h/2\pi$

Discrete levels, $E_n \propto n^2$. For a box a centimetre long they are so close they form a near-continuum.

$E_n \propto n^2$



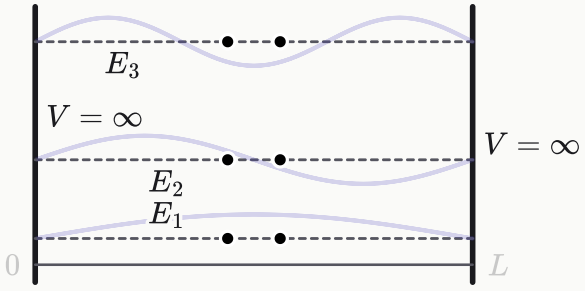
FILL THE LEVELS

$$2n_F = N$$

Pauli: two electrons (spin up, spin down) per level

Pauli exclusion principle (taken as a law of nature): no two electrons share a state, and each level n has two spin states. At 0 K the N electrons fill the lowest levels two at a time. The top filled level n_F is the Fermi level (one dimension first).

Two electrons per level



Check. How many electrons can one level n hold?

- ✓ 2, one of each spin
- 1
- n
- $2n^2$

Each n has two spin states, $m_s = \pm\frac{1}{2}$, and Pauli allows one electron per state.

$$E_F = \frac{h^2}{8mL^2}\left(\frac{N}{2}\right)^2$$

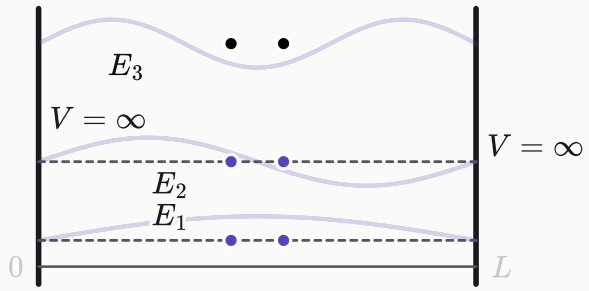
substitute: $n = n_F = N/2$

$$E_F = \frac{h^2 N^2}{32mL^2}$$

simplify

E_F depends only on the line density N/L. With $N/L = 0.5$ electron per Å, $E_F \approx 2.4$ eV. The average energy per electron in 1-D is one third of E_F (the 1-D case).

The top filled level is E_F



Fermi energy of a 3-D electron gas

5–8 marks

Cube of side L, volume $V = L^3$, periodic boundary conditions. Allowed k_x, k_y, k_z are multiples of $2\pi/L$. N electrons, density $n = N/V$. k_F: radius of the filled sphere at 0 K.

BEFORE YOU START: WHAT THIS USES

$\psi_{\mathbf{k}} \propto e^{i\mathbf{k}\cdot\mathbf{r}}$	free-electron plane wave in a box: three dimensions and the Fermi sphere
$E = \hbar^2 k^2/2m$	kinetic energy from $p = \hbar k$: first steps of the 1-D box
Pauli	two electrons (spin up and down) per allowed $\mathbf{k}$: stated in the 1-D box
$\frac{4}{3}\pi k^3$	volume of a sphere in k-space: derived in the step "Volume of a sphere" below

START

$$k_x = \frac{2\pi n_x}{L}$$

RESULT

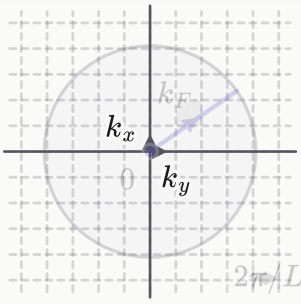
$$E_F = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}$$

ALLOWED STATES

$$k_x = \frac{2\pi n_x}{L}$$

A travelling wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ that repeats after L in each direction needs $e^{ik_xL} = 1$, so $k_x = 2\pi n_x/L$ with $n_x = 0, \pm1, \pm2, \ldots$, and the same for k_y, k_z. (Periodic boundary conditions are bookkeeping: for a large crystal the count of states does not depend on the choice of boundary condition.)

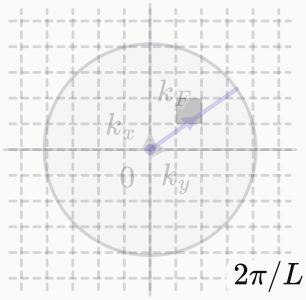
Grid of allowed k, spacing $2\pi/L$



$$\text{volume per state} = \left(\frac{2\pi}{L}\right)^3$$

Each allowed $\mathbf{k}$ owns one small cube of $\mathbf{k}$-space

One state per $(2\pi/L)^3$



VOLUME OF A SPHERE

$$V_{\text{sphere}} = \int_{-k_F}^{k_F} \pi\left(k_F^2 - k_z^2\right) dk_z$$

Slice the sphere into discs perpendicular to k_z

At height k_z the slice is a disc of radius $\sqrt{k_F^2 - k_z^2}$, area $\pi(k_F^2 - k_z^2)$.

$$V_{\text{sphere}} = \pi\left[k_F^2 k_z - \frac{k_z^3}{3}\right]_{-k_F}^{k_F} = \tfrac{4}{3}\pi k_F^3$$

evaluate

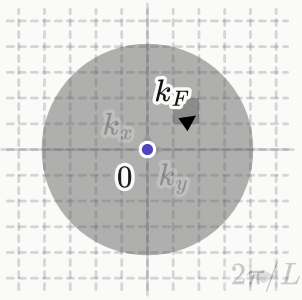
COUNT THE FILLED SPHERE

$$N = 2 \cdot \frac{\frac{4}{3}\pi k_F^3}{(2\pi/L)^3}$$

States in the sphere = sphere volume ÷ volume per state, $\times 2$ for spin

$E = \hbar^2 k^2/2m$ (1-D box), so all states of one energy lie on a sphere. At 0 K they fill a sphere of radius k_F.

Count the k-points inside radius k_F



Check. Why the factor 2?

- ✓ two spin states for each $\mathbf{k}$
- the sphere has two halves
- $\pm\boldsymbol{k}$ give the same energy
- it comes from $\hbar = h/2\pi$

Each allowed $\mathbf{k}$ can hold one spin-up and one spin-down electron.

$$N = \frac{V}{3\pi^2} k_F^3$$

simplify

$2 \cdot \frac{4}{3}\pi \cdot L^3/(8\pi^3) = L^3/3\pi^2$.

$$k_F^3 = 3\pi^2 \frac{N}{V}$$

multiply: $\frac{3\pi^2}{V}$

$$k_F = \left(3\pi^2 n\right)^{1/3}$$

root: 3

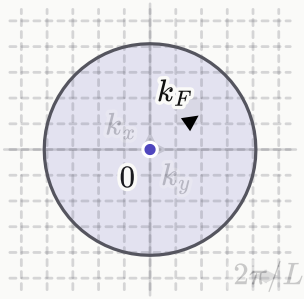
k_F depends only on the electron density $n = N/V$, not on the size or shape of the crystal.

FERMI ENERGY

$$E_F = \frac{\hbar^2 k_F^2}{2m}$$

Energy at the surface of the sphere

E_F at radius k_F



$$E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$$

substitute: $k_F = (3\pi^2 n)^{1/3}$

An equivalent way to write this is $E_F = \dfrac{h^2}{8m}\left(\dfrac{3n}{\pi}\right)^{2/3}$. Then the Fermi speed is $v_F = \hbar k_F / m$ and the Fermi temperature $T_F = E_F / k_B$ (Fermi parameters). Metals: $E_F \approx 2$–7 eV, $T_F \sim 10^4$–10^5 K.

Density of states $D(E)$ in 3-D

5 marks

$N(E)$: number of electron states with energy less than E. $D(E) = dN/dE$: number of states per unit energy range. Cube of volume V.

BEFORE YOU START: WHAT THIS USES

$N(E)$	number of electron states with energy below E: the 3-D Fermi-energy derivation
$D(E) = dN/dE$	states per unit energy: what $D(E)$ means
$k = \sqrt{2mE}/\hbar$	free-electron relation $E = \hbar^2 k^2/2m$: 1-D box

START

$$N(E) = \frac{V}{3\pi^2}k^3$$

RESULT

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

STATES BELOW E

$$N(E) = \frac{V}{3\pi^2}k^3$$

The count in the Fermi-energy derivation holds for a sphere of any radius k, not just k_F.

$$N(E) = \frac{V}{3\pi^2}\left(\frac{2mE}{\hbar^2}\right)^{3/2}$$

substitute: $k = \sqrt{2mE}/\hbar$

DIFFERENTIATE

$$D(E) = \frac{dN}{dE}$$

Density of states

Check. $N \propto E^{3/2}$. How does $D = dN/dE$ depend on E?

- ✓ $D \propto E^{1/2}$
- $D \propto E^{3/2}$
- $D \propto E^{-1/2}$
- D is constant

$\frac{d}{dE}E^{3/2} = \frac{3}{2}E^{1/2}$.

$$D(E) = \frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \cdot \frac{3}{2}E^{1/2}$$

differentiate with respect to E

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$$

simplify

In the usual units (per unit volume, with h): $D(E) = \dfrac{4\pi}{h^3}(2m)^{3/2}E^{1/2}$. It rises as $\sqrt{E}$. In 1-D the same steps give $D \propto E^{-1/2}$; in 2-D, a constant (why the dimension matters).

AT THE FERMI LEVEL

$$D(E_F) = \frac{3N}{2E_F}$$

Divide D *by* $N = \frac{2}{3}E_F D(E_F)$

Since $N \propto E_F^{3/2}$, $dN/dE = \frac{3}{2}N/E$ at E_F. A quick result for numericals.

THE SAME WITH h

$$D(E) = \frac{4\pi V}{h^3}(2m)^{3/2}E^{1/2}$$

substitute: $\hbar = h/2\pi$

$\hbar^{-3} = (2\pi)^3/h^3$ and $\frac{(2\pi)^3}{2\pi^2} = 4\pi$. Divide by V for the density of states per unit volume, the form used for semiconductors in Module III.

Average energy at 0 K is $\frac{3}{5}E_F$

4 marks

$D(E) = CE^{1/2}$ with $C = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2}$. At 0 K every state below E_F is filled and every state above is empty.

BEFORE YOU START: WHAT THIS USES

$f(E)$	Fermi function; at 0 K it is 1 below E_F and 0 above: the Fermi function
$D(E)$	density of states, $D = CE^{1/2}$: density-of-states derivation
E_F	Fermi energy: Fermi-energy derivation

START

$$E_{\text{tot}} = \int_0^{E_F} E\, D(E)\, dE$$

RESULT

$$\bar{E}_0 = \frac{3}{5} E_F$$

TOTAL ENERGY

$$E_{\text{tot}} = \int_0^{E_F} E\, D(E)\, dE$$

At 0 K, $f(E) = 1$ below E_F, 0 above. Each filled state contributes its energy E.

$$E_{\text{tot}} = C \int_0^{E_F} E^{3/2}\, dE = \frac{2}{5} C E_F^{5/2}$$

substitute: $D = CE^{1/2}$

NUMBER OF ELECTRONS

$$N = \int_0^{E_F} D(E)\, dE = \frac{2}{3} C E_F^{3/2}$$

Count the filled states the same way

DIVIDE

$$\bar{E}_0 = \frac{E_{\text{tot}}}{N} = \frac{\frac{2}{5} C E_F^{5/2}}{\frac{2}{3} C E_F^{3/2}}$$

Average energy per electron

Check. What is $\dfrac{2/5}{2/3}$?

- ✓ $3/5$
- $4/15$
- $5/3$
- $1/2$

$\frac{2}{5} \times \frac{3}{2} = \frac{3}{5}$.

$$\bar{E}_0 = \frac{3}{5} E_F$$

simplify

Above the midpoint $\frac{1}{2} E_F$, because $D \propto \sqrt{E}$ puts more states near the top. Even at 0 K the gas has a large kinetic energy (Pauli forbids everyone sitting at $E = 0$). The degeneracy pressure and the bulk modulus follow in the steps below.

PRESSURE AND BULK MODULUS

$$P = -\frac{\partial E_{\text{tot}}}{\partial V}$$

Compressing the gas costs energy: work done on it is $-P\, dV$

At 0 K and fixed N, so no heat is exchanged.

$$E_{\text{tot}} = \tfrac{3}{5} N E_F \propto V^{-2/3}$$

substitute: $E_F \propto (N/V)^{2/3}$

From $E_F = \frac{\hbar^2}{2m} (3\pi^2 N/V)^{2/3}$ (Fermi-energy derivation).

$$P = \frac{2}{3} \frac{E_{\text{tot}}}{V}$$

differentiate with respect to V

$-\frac{d}{dV} V^{-2/3} = \frac{2}{3} V^{-5/3}$.

$$B = -V \frac{dP}{dV} = \frac{5}{3} P$$

Bulk modulus: pressure change per fractional volume change

$P \propto V^{-5/3}$, so $-V\, dP/dV = \frac{5}{3} P$.

Electronic specific heat: only a fraction T/T_F can be excited

5 marks

$T_F = E_F/k_B$ (Fermi temperature). Classically every electron would take $\frac{3}{2} k_B T$.

BEFORE YOU START: WHAT THIS USES

$T_F = E_F/k_B$	Fermi temperature: Fermi parameters
$k_B T$	thermal energy; the Fermi function changes from 1 to 0 over about this width: the Fermi function
$E = \frac{3}{2} N k_B T$	classical equipartition: derived in the first steps below
Pauli	an electron can only move into an empty state: Pauli principle

START

$$\left\langle \tfrac{1}{2} m v_x^2 \right\rangle = \frac{\int \frac{1}{2} m v_x^2\, e^{-m v_x^2/2k_BT}\, dv_x}{\int e^{-m v_x^2/2k_BT}\, dv_x} = \tfrac{1}{2} k_B T$$

RESULT

$$C_v = \frac{3}{2} N k_B \frac{T}{T_F}$$

THE CLASSICAL ANSWER

$$\left\langle \tfrac{1}{2} m v_x^2 \right\rangle = \frac{\int \frac{1}{2} m v_x^2\, e^{-m v_x^2/2k_BT}\, dv_x}{\int e^{-m v_x^2/2k_BT}\, dv_x} = \tfrac{1}{2} k_B T$$

Average over the Boltzmann distribution (the classical gas, Drude–Lorentz)

With $a = m/2k_BT$, $\int v^2 e^{-av^2} dv = \frac{1}{2a} \int e^{-av^2} dv$, so the ratio is $\frac{1}{2} m \cdot \frac{1}{2a} = \frac{1}{2} k_B T$.

$$E = \frac{3}{2} N k_B T$$

Three independent directions x, y, z, N *electrons*

Each electron has $3 \times \frac{1}{2} k_B T$: classical (Drude–Lorentz) theory.

$$C_v = \frac{3}{2} N k_B$$

differentiate with respect to T

About $\frac{3}{2} R$ per mole of electrons. Experiment shows about 1% of this, growing in proportion to T.

WHO CAN ACTUALLY TAKE HEAT

$$N_{\text{eff}} = N\frac{k_BT}{E_F}$$

Only electrons within about k_BT of E_F have empty states to move into

An electron deep in the Fermi sea would have to jump into a state that is already full. Only those within about k_BT below E_F can gain $\sim k_BT$: that is the width over which $f(E)$ falls from 1 to 0 (the Fermi function).

Check. Why can't an electron 1 eV below E_F absorb $k_BT \approx 0.025$ eV?

- ✓ the states 0.025 eV above it are all occupied
- it is bound to an ion
- it has no kinetic energy
- k_BT is too large

Pauli: it can only move into an empty state, and the only empty states are above E_F.

$$N_{\text{eff}} = N\frac{T}{T_F}$$

substitute: $E_F = k_BT_F$

QUANTUM ANSWER

$$C_v = \frac{3}{2}N_{\text{eff}}k_B = \frac{3}{2}Nk_B\frac{T}{T_F}$$

Use N_{eff} in place of N

With $T = 300$ K and $T_F \sim 3\times10^4$ K this is about 1% of the classical value, and it is proportional to T, as observed. (The exact Sommerfeld result is $\frac{\pi^2}{2}Nk_BT/T_F$.)

Bloch theorem: $\psi(x+a) = e^{ika}\psi(x)$ 5–8 marks

$V(x+a) = V(x)$: a periodic potential with lattice constant a. $f(x)$, $g(x)$: two real independent solutions of the Schrödinger equation. λ: a constant.

BEFORE YOU START: WHAT THIS USES

$V(x+a) = V(x)$	the potential of a perfect crystal repeats with the lattice period a: where the gaps come from
Schrödinger equation	$\psi'' = -\frac{2m}{\hbar^2}(E-V)\psi$: how it arises in the 1-D box
f, g	two independent solutions: a second-order linear equation has exactly two
$W = fg' - gf'$	the Wronskian: shown constant in the steps below
λ	a constant multiplier, found from the quadratic below

START

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0$$

RESULT

$$\psi(x) = e^{ikx}u_k(x)$$

SHIFT BY ONE LATTICE STEP

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0$$

The equation does not change when $x \to x + a$, because $V(x+a) = V(x)$. So $f(x+a)$ and $g(x+a)$ are also solutions.

$$f(x+a) = \alpha_1 f(x) + \alpha_2 g(x)$$

A second-order equation has only two independent solutions

and $g(x+a) = \beta_1 f(x) + \beta_2 g(x)$.

PICK THE SPECIAL COMBINATION

$$\psi(x+a) = \lambda\psi(x)$$

Choose $\psi = Af + Bg$ so that a shift only multiplies it

This needs $A\alpha_1 + B\beta_1 = \lambda A$, $A\alpha_2 + B\beta_2 = \lambda B$, which has a non-zero solution only if the determinant vanishes.

$$(\alpha_1 - \lambda)(\beta_2 - \lambda) - \alpha_2\beta_1 = 0$$

Determinant of the two conditions

Matching the coefficients of f and g in $A(\alpha_1 f + \alpha_2 g) + B(\beta_1 f + \beta_2 g) = \lambda(Af + Bg)$ gives $A\alpha_1 + B\beta_1 = \lambda A$ and $A\alpha_2 + B\beta_2 = \lambda B$.

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + (\alpha_1\beta_2 - \alpha_2\beta_1) = 0$$

expand

$$\frac{dW}{dx} = fg'' - gf'' = 0$$

The Wronskian $W = fg' - gf'$ is constant

Both f and g obey $\psi'' = -\frac{2m}{\hbar^2}(E-V)\psi$, so $fg'' - gf'' = -\frac{2m}{\hbar^2}(E-V)(fg - gf) = 0$.

$$W(x+a) = (\alpha_1\beta_2 - \alpha_2\beta_1)\,W(x)$$

substitute: $f(x+a)$, $g(x+a)$ from the first lines

Expand $(\alpha_1 f + \alpha_2 g)(\beta_1 f' + \beta_2 g') - (\beta_1 f + \beta_2 g)(\alpha_1 f' + \alpha_2 g')$: the ff' and gg' terms cancel and $(\alpha_1\beta_2 - \alpha_2\beta_1)(fg' - gf')$ is left.

$$\alpha_1\beta_2 - \alpha_2\beta_1 = 1$$

W is constant, so $W(x+a) = W(x)$

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + 1 = 0$$

Put the constant term into the quadratic

The product of the roots is 1.

THE ROOTS

$$\lambda_1 \lambda_2 = 1$$

Product of the roots of the quadratic

Check. The roots multiply to 1, and the solution must stay finite over the whole (huge) crystal. What must λ be?

- ✓ $e^{\pm ika}$ with real k, so $|\lambda| = 1$
- any real number
- $\lambda > 1$
- $\lambda = 0$

If $|\lambda| > 1$, ψ grows without limit after many steps; if $|\lambda| < 1$, the other root does. Only $|\lambda| = 1$ works: these energies form the allowed bands.

$$\psi(x + a) = e^{ika}\psi(x)$$

substitute: $\lambda = e^{ika}$

When $(\alpha_1 + \beta_2)^2 < 4$ the roots are complex conjugates on the unit circle. Energies with $(\alpha_1 + \beta_2)^2 > 4$ have no bounded solution: forbidden gaps.

BLOCH FORM

$$\psi(x) = e^{ikx}u_k(x), \quad u_k(x + a) = u_k(x)$$

Check: this form has exactly that property

$\psi(x + a) = e^{ik(x+a)}u_k(x + a) = e^{ika}e^{ikx}u_k(x) = e^{ika}\psi(x)$. A plane wave modulated by a function with the lattice period: a Bloch function.

The Kronig–Penney model

8 marks

Wells of width a ($V = 0$) between barriers of width b and height V_0; period $a + b$. $E < V_0$. $\alpha^2 = 2mE/\hbar^2$, $\beta^2 = 2m(V_0 - E)/\hbar^2$.

BEFORE YOU START: WHAT THIS USES

V_0, a, b	barrier height, well width, barrier width: the model and its one equation
α, β	wave number in the well and decay constant in the barrier, from $\psi'' = -\frac{2m}{\hbar^2}(E - V)\psi$: Schrödinger equation
Bloch	one period later ψ is multiplied by $e^{ik(a+b)}$: Bloch's theorem
$E < V_0$	bound-electron case, energies below the barrier top
P	barrier strength: defined in the step "Thin, high barriers"

START

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad (0 < x < a)$$

RESULT

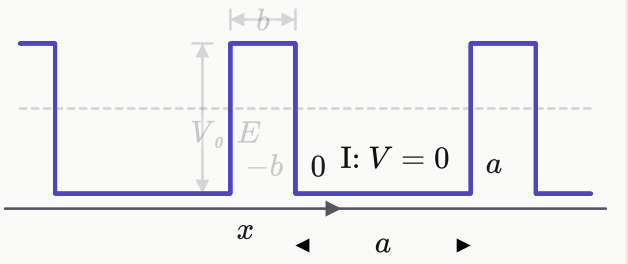
$$P\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

TWO REGIONS

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \quad (0 < x < a)$$

Inside a well $V = 0$.

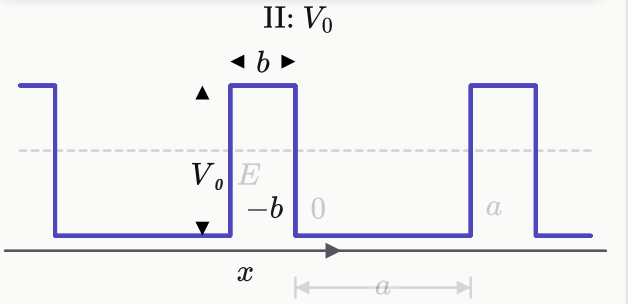
Region I, the well



$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \quad (-b < x < 0)$$

Inside a barrier, $V = V_0 > E$

Region II, the barrier

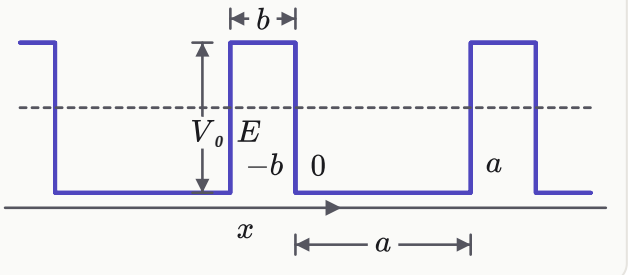


SOLVE IN EACH REGION

$$\psi(x) = \psi(0)\cos \alpha x + \psi'(0)\frac{\sin \alpha x}{\alpha}$$

General solution of $\psi'' = -\alpha^2\psi$*, written with its starting values*

$\cos \alpha x$ and $\sin \alpha x/\alpha$ have (value, slope) $= (1, 0)$ and $(0, 1)$ at $x = 0$, so any solution is a mix of them with the weights $\psi(0)$ and $\psi'(0)$.



$$\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_a = \begin{pmatrix}\cos\alpha a & \dfrac{\sin\alpha a}{\alpha}\\ -\alpha\sin\alpha a & \cos\alpha a\end{pmatrix}\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_0$$

Carry (ψ,ψ') across the well: matrix M_w

The second row is the derivative of the solution above.

$$\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_b = \begin{pmatrix}\cosh\beta b & \dfrac{\sinh\beta b}{\beta}\\ \beta\sinh\beta b & \cosh\beta b\end{pmatrix}\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_0$$

Across the barrier, $\psi''=+\beta^2\psi$: matrix M_b

Now the two basic solutions are $\cosh\beta x$ and $\sinh\beta x/\beta$.

JOIN THE REGIONS

$$\psi,\ \psi' \text{ continuous at each boundary}$$

Why ψ and ψ' match

The Schrödinger equation makes ψ'' finite, so ψ' has no jump, and then ψ has none either. So the barrier matrix can be applied straight after the well matrix.

$$M = M_b M_w$$

One full period: well, then barrier

$$\det M = \det M_b\ \det M_w = 1$$

Each matrix has determinant 1

$\det M_w = \cos^2\alpha a + \sin^2\alpha a = 1$ and
$\det M_b = \cosh^2\beta b - \sinh^2\beta b = 1$.

BLOCH'S CONDITION

$$\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_{a+b} = e^{ik(a+b)}\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_0$$

Bloch's theorem: one period later ψ is multiplied by $e^{ik(a+b)}$

The same factor applies to the slope. See the Bloch derivation.

$$M\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_0 = \lambda\begin{pmatrix}\psi\\ \psi'\end{pmatrix}_0,\quad \lambda = e^{ik(a+b)}$$

So λ is an eigenvalue of M

$$\lambda^2 - (\operatorname{Tr} M)\,\lambda + \det M = 0$$

Eigenvalue equation of a 2×2 matrix

$$\lambda + \frac{1}{\lambda} = \operatorname{Tr} M$$

Divide by λ, using $\det M = 1$

$$\cos k(a+b) = \tfrac{1}{2}\operatorname{Tr} M$$

substitute: $\lambda+\lambda^{-1} = 2\cos k(a+b)$

THE TRACE

$$\operatorname{Tr} M = 2\cosh\beta b\cos\alpha a + \left(\frac{\beta}{\alpha}-\frac{\alpha}{\beta}\right)\sinh\beta b\,\sin\alpha a$$

Multiply $M_b M_w$ and add the two diagonal entries

The diagonal entries are $\cosh\beta b\cos\alpha a - \frac{\alpha}{\beta}\sinh\beta b\sin\alpha a$ and $\frac{\beta}{\alpha}\sinh\beta b\sin\alpha a + \cosh\beta b\cos\alpha a$.

$$\frac{\beta^2-\alpha^2}{2\alpha\beta}\sinh\beta b\,\sin\alpha a + \cosh\beta b\,\cos\alpha a = \cos k(a+b)$$

$$\tfrac{\beta}{\alpha} - \tfrac{\alpha}{\beta} = \tfrac{\beta^2-\alpha^2}{\alpha\beta}$$

The exact Kronig–Penney condition, with $Q = \frac{\beta^2-\alpha^2}{2\alpha\beta}$. The usual textbook route (a 4×4 determinant in A, B, C, D) is the same algebra written differently.

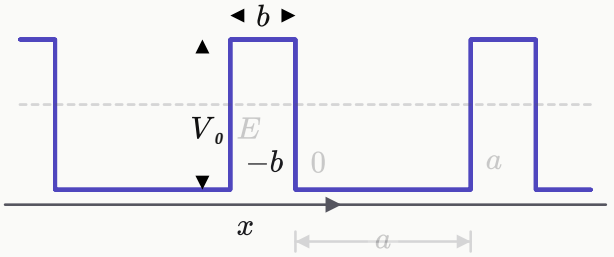
THIN, HIGH BARRIERS

$$P = \frac{mV_0 ba}{\hbar^2}$$

Let $V_0\to\infty$, $b\to 0$, keeping $V_0 b$ finite

P measures the barrier strength (its area $V_0 b$): how tightly an electron is held in one well. With $b\to 0$: $\sinh\beta b\to\beta b$, $\cosh\beta b\to 1$, $\beta^2\gg\alpha^2$.

Barrier area $V_0 b$ stays finite



Check. As $b\to 0$, what do $\sinh\beta b$ and $\cosh\beta b$ become?

- ✓ βb and 1
- 0 and 0
- 1 and βb
- βb and βb

For a small argument $\sinh x\approx x$ and $\cosh x\approx 1$.

$$\frac{\beta^2-\alpha^2}{2\alpha\beta}\sinh\beta b\,\sin\alpha a \ \to\ \frac{\beta^2 b}{2\alpha}\sin\alpha a$$

$\sinh\beta b\to\beta b$ *and* $\beta^2\gg\alpha^2$

$$\frac{\beta^2 b}{2\alpha} = \frac{mV_0 b}{\hbar^2\alpha} = \frac{P}{\alpha a}$$

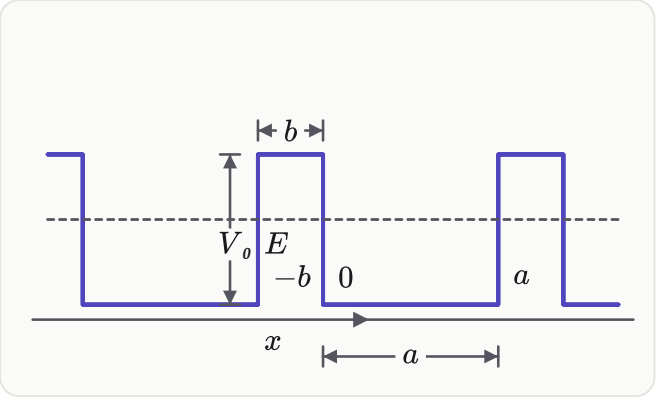
substitute: $\beta^2\approx 2mV_0/\hbar^2$

For $V_0\gg E$, $\beta^2 = 2m(V_0-E)/\hbar^2\approx 2mV_0/\hbar^2$. Multiplying top and bottom by a gives $P/\alpha a$ with $P = mV_0 ba/\hbar^2$.

$$P\frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka$$

simplify

The Kronig–Penney condition. Also $a + b \to a$ because $b \to 0$.



WHAT IT MEANS

$$-1 \le P\frac{\sin \alpha a}{\alpha a} + \cos \alpha a \le 1$$

The right side is a cosine

Only energies (values of αa) that keep the left side between -1 and 1 are allowed: **allowed bands separated by forbidden gaps**. Bands widen as the energy rises and narrow as P grows.

$$P \to 0: \quad \alpha = k, \quad E = \frac{\hbar^2 k^2}{2m}$$

evaluate: $P \to 0$

No barriers: free electrons, no gaps ($E = \hbar^2 k^2/2m$ as in the 3-D Fermi-energy derivation).

$$P \to \infty: \quad \sin \alpha a = 0, \quad E = \frac{n^2 h^2}{8ma^2}$$

evaluate: $P \to \infty$

Impenetrable barriers: each well is an isolated box, and the bands shrink to the discrete levels of a particle in a box of width a, as derived in the 1-D box.

Velocity of a Bloch electron: $v = \frac{1}{\hbar}\frac{dE}{dk}$

3 marks

An electron in a band is a wave packet; it moves with the group velocity $v_g = d\omega/dk$. $E = \hbar\omega$.

BEFORE YOU START: WHAT THIS USES

$\omega = E/\hbar$	Planck–Einstein relation, also true for matter waves
$E(k)$	band energy of a Bloch state: Bloch's theorem
$v_g = d\omega/dk$	group velocity: derived in the first two steps below
wave packet	a localised electron is a sum of waves with nearby k

START

$$\cos(kx - \omega t) + \cos\left((k+\delta k)x - (\omega + \delta\omega)t\right) = 2\cos\tfrac{1}{2}(\delta k\, x - \delta\omega\, t)\ \cos(\text{fast wave})$$

RESULT

$$v = \frac{1}{\hbar}\frac{dE}{dk}$$

TWO NEARBY WAVES

$$\cos(kx - \omega t) + \cos\left((k+\delta k)x - (\omega + \delta\omega)t\right) = 2\cos\tfrac{1}{2}(\delta k\, x - \delta\omega\, t)\ \cos(\text{fast wave})$$

Two nearby waves beat: the sum has a slowly varying envelope

$\cos P + \cos Q = 2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$. The factor $\cos\frac{1}{2}(\delta k\, x - \delta\omega\, t)$ is the envelope, a wave packet.

$$\delta k\, x - \delta\omega\, t = \text{const} \quad \Rightarrow \quad v = \frac{x}{t} = \frac{\delta\omega}{\delta k}$$

Follow a point of the envelope

This is the speed of the envelope, which carries the electron, not the speed ω/k of the crests inside it.

GROUP VELOCITY

$$v = \frac{d\omega}{dk}$$

Let $\delta k \to 0$: a localised electron is a packet of Bloch waves with k near some value, and it moves at the group velocity $d\omega/dk$.

$$v = \frac{d(E/\hbar)}{dk}$$

substitute: $\omega = E/\hbar$

The energy of a wave is $E = \hbar\omega$ (Planck–Einstein, also for matter waves).

$$v = \frac{1}{\hbar}\frac{dE}{dk}$$

simplify

The slope of the band. For free electrons $E = \hbar^2 k^2/2m$ gives $v = \hbar k/m$, as expected. At the bottom and top of a band and at the zone boundary $k = \pm\pi/a$, $dE/dk = 0$: the electron does not move (a standing wave).

Effective mass $m^* = \hbar^2/(d^2E/dk^2)$

5 marks

$\mathcal{E}$: applied field. $v = \frac{1}{\hbar}\frac{dE}{dk}$ (group velocity). The electron's charge is taken as e in magnitude; only the form of the result matters here.

BEFORE YOU START: WHAT THIS USES

$v = \frac{1}{\hbar}\frac{dE}{dk}$	group velocity of a Bloch electron: group-velocity derivation
$F = e\mathcal{E}$	force on a charge e in a field $\mathcal{E}$
$dE = F\,v\,dt$	work = force × distance moved in dt
m^*	effective mass: speed and effective mass
$a = F/m^*$	the comparison with Newton's second law made in the last step

START

$$dE = e\mathcal{E}\,v\,dt$$

RESULT

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

WORK DONE BY THE FIELD

$$dE = e\mathcal{E}\,v\,dt$$

In a time dt the field does work force × distance $= e\mathcal{E}\cdot v\,dt$.

$$dE = \frac{e\mathcal{E}}{\hbar}\frac{dE}{dk}\,dt$$

substitute: $v = \frac{1}{\hbar}\frac{dE}{dk}$

$$\frac{dE}{dk}dk = \frac{e\mathcal{E}}{\hbar}\frac{dE}{dk}\,dt$$

substitute: $dE = \frac{dE}{dk}dk$

$$\hbar\frac{dk}{dt} = e\mathcal{E}$$

cancel

Cancel dE/dk and rearrange. $\hbar k$ behaves like momentum (crystal momentum) and changes at the rate of the external force.

ACCELERATION

$$a = \frac{dv}{dt} = \frac{1}{\hbar}\frac{d^2E}{dk^2}\frac{dk}{dt}$$

differentiate with respect to t

Chain rule: v depends on t through k.

Check. Using $\hbar\,dk/dt = e\mathcal{E}$, what does dk/dt become?

- ✓ $e\mathcal{E}/\hbar$
- $\hbar e\mathcal{E}$
- $e\mathcal{E}/m$
- $\hbar/e\mathcal{E}$

Divide both sides by $\hbar$.

$$a = \frac{e\mathcal{E}}{\hbar^2}\frac{d^2E}{dk^2}$$

substitute: $\frac{dk}{dt} = \frac{e\mathcal{E}}{\hbar}$

COMPARE WITH NEWTON

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

Compare with $a = F/m^*$, $F = e\mathcal{E}$

The curvature of the band plays the part of mass. Bottom of a band: $m^* > 0$. Top of a band: $m^* < 0$ (the electron behaves like a positive charge, a hole: holes). At the inflection point $m^* \to \infty$. For free electrons $d^2E/dk^2 = \hbar^2/m$ gives $m^* = m$.